

理论物理笔记

Theoretical Physics Notes

Yee

一份简明的物理学手册编纂计划

Theoretical Physics Notes

First Edition

Yee

School of Physics, WHU

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Preface

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Chapter 1 向量

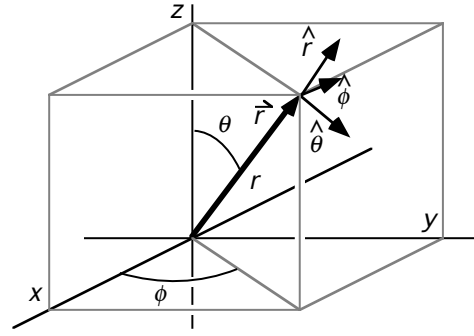
Spherical Coordinates

Transforms

The forward and reverse coordinate transformations are

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} & x &= r \sin \theta \cos \phi \\ \theta &= \arctan\left(\sqrt{x^2 + y^2}, z\right) & y &= r \sin \theta \sin \phi \\ \phi &= \arctan(y, x) & z &= r \cos \theta \end{aligned}$$

where we *formally* take advantage of the *two argument* arctan function to eliminate quadrant confusion.



Unit Vectors

The unit vectors in the spherical coordinate system are functions of position. It is convenient to express them in terms of the *spherical* coordinates and the unit vectors of the *rectangular* coordinate system which are *not* themselves functions of position.

$$\begin{aligned} \hat{r} &= \frac{\vec{r}}{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{r} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta \\ \hat{\phi} &= \frac{\hat{z} \times \hat{r}}{\sin \theta} = -\hat{x} \sin \phi + \hat{y} \cos \phi \\ \hat{\theta} &= \hat{\phi} \times \hat{r} = \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta \end{aligned}$$

Variations of unit vectors with the coordinates

Using the expressions obtained above it is easy to derive the following handy relationships:

$$\begin{aligned} \frac{\partial \hat{r}}{\partial r} &= 0 \\ \frac{\partial \hat{r}}{\partial \theta} &= \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta = \hat{\theta} \\ \frac{\partial \hat{r}}{\partial \phi} &= -\hat{x} \sin \theta \sin \phi + \hat{y} \sin \theta \cos \phi = (-\hat{x} \sin \phi + \hat{y} \cos \phi) \sin \theta = \hat{\phi} \sin \theta \\ \frac{\partial \hat{\phi}}{\partial r} &= 0 \\ \frac{\partial \hat{\phi}}{\partial \theta} &= 0 \\ \frac{\partial \hat{\phi}}{\partial \phi} &= -\hat{x} \cos \phi - \hat{y} \sin \phi = -(\hat{r} \sin \theta + \hat{\theta} \cos \theta) \\ \frac{\partial \hat{\theta}}{\partial r} &= 0 \\ \frac{\partial \hat{\theta}}{\partial \theta} &= -\hat{x} \sin \theta \cos \phi - \hat{y} \sin \theta \sin \phi - \hat{z} \cos \theta = -\hat{r} \\ \frac{\partial \hat{\theta}}{\partial \phi} &= -\hat{x} \cos \theta \sin \phi + \hat{y} \cos \theta \cos \phi = \hat{\phi} \cos \theta \end{aligned}$$

Path increment

We will have many uses for the path increment $d\vec{r}$ expressed in spherical coordinates:

$$\begin{aligned} d\vec{r} &= d(r\hat{r}) = \hat{r}dr + r d\hat{r} = \hat{r}dr + r \left(\frac{\partial \hat{r}}{\partial r} dr + \frac{\partial \hat{r}}{\partial \theta} d\theta + \frac{\partial \hat{r}}{\partial \phi} d\phi \right) \\ &= \hat{r}dr + \hat{\theta}r d\theta + \hat{\phi}r \sin \theta d\phi \end{aligned}$$

Time derivatives of the unit vectors

We will also have many uses for the time derivatives of the unit vectors expressed in spherical coordinates:

$$\begin{aligned} \dot{\hat{r}} &= \frac{\partial \hat{r}}{\partial r} \dot{r} + \frac{\partial \hat{r}}{\partial \theta} \dot{\theta} + \frac{\partial \hat{r}}{\partial \phi} \dot{\phi} = \hat{\theta} \dot{\theta} + \hat{\phi} \dot{\phi} \sin \theta \\ \dot{\hat{\theta}} &= \frac{\partial \hat{\theta}}{\partial r} \dot{r} + \frac{\partial \hat{\theta}}{\partial \theta} \dot{\theta} + \frac{\partial \hat{\theta}}{\partial \phi} \dot{\phi} = -\hat{r} \dot{\theta} + \hat{\phi} \dot{\phi} \cos \theta \\ \dot{\hat{\phi}} &= \frac{\partial \hat{\phi}}{\partial r} \dot{r} + \frac{\partial \hat{\phi}}{\partial \theta} \dot{\theta} + \frac{\partial \hat{\phi}}{\partial \phi} \dot{\phi} = -(\hat{r} \sin \theta + \hat{\theta} \cos \theta) \dot{\phi} \end{aligned}$$

Velocity and Acceleration

The velocity and acceleration of a particle may be expressed in spherical coordinates by taking into account the associated rates of change in the unit vectors:

$$\begin{aligned} \vec{v} &= \dot{\vec{r}} = \dot{r}\hat{r} + \hat{r}\dot{r} \\ \vec{v} &= \hat{r}\dot{r} + \hat{\theta}r\dot{\theta} + \hat{\phi}r\dot{\phi}\sin\theta \\ \vec{a} &= \dot{\vec{v}} = \dot{\hat{r}}\dot{r} + \hat{r}\ddot{r} + \dot{\hat{\theta}}r\dot{\theta} + \hat{\theta}r\ddot{\theta} + \dot{\hat{\theta}}\dot{\theta} + \hat{\theta}r\ddot{\theta} + \dot{\hat{\phi}}r\dot{\phi}\sin\theta + \hat{\phi}r\ddot{\phi}\sin\theta + \hat{\phi}r\dot{\phi}\dot{\theta}\cos\theta + \hat{\phi}r\dot{\phi}\dot{\theta}\cos\theta \\ &= (\hat{\theta}\dot{\theta} + \hat{\phi}\dot{\phi}\sin\theta)\dot{r} + \hat{r}\ddot{r} + (-\hat{r}\dot{\theta} + \hat{\phi}\dot{\phi}\cos\theta)r\dot{\theta} + \hat{\theta}r\ddot{\theta} + \hat{\theta}r\dot{\theta} \\ &\quad + \left[-(\hat{r}\sin\theta + \hat{\theta}\cos\theta)\dot{\phi} \right] r\dot{\phi}\sin\theta + \hat{\phi}r\ddot{\phi}\sin\theta + \hat{\phi}r\dot{\phi}\dot{\theta}\cos\theta + \hat{\phi}r\dot{\phi}\dot{\theta}\cos\theta \\ \vec{a} &= \hat{r}(\ddot{r} - r\dot{\theta}^2 - r\dot{\phi}^2 \sin^2 \theta) + \hat{\theta}(r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\dot{\phi}^2 \sin \theta \cos \theta) + \hat{\phi}(r\ddot{\phi} \sin \theta + 2r\dot{\theta}\dot{\phi} \cos \theta + 2\dot{r}\dot{\phi} \sin \theta) \end{aligned}$$

The del operator from the definition of the gradient

Any (static) scalar field u may be considered to be a function of the spherical coordinates r , θ , and ϕ . The value of u changes by an infinitesimal amount du when the point of observation is changed by $d\vec{r}$. That change may be determined from the partial derivatives as

$$du = \frac{\partial u}{\partial r} dr + \frac{\partial u}{\partial \theta} d\theta + \frac{\partial u}{\partial \phi} d\phi.$$

But we also define the gradient in such a way as to obtain the result

$$du = \vec{\nabla} u \cdot d\vec{r}$$

Therefore,

$$\frac{\partial u}{\partial r} dr + \frac{\partial u}{\partial \theta} d\theta + \frac{\partial u}{\partial \phi} d\phi = \vec{\nabla} u \cdot d\vec{r}$$

or, in spherical coordinates,

$$\frac{\partial u}{\partial r} dr + \frac{\partial u}{\partial \theta} d\theta + \frac{\partial u}{\partial \phi} d\phi = (\vec{\nabla} u)_r dr + (\vec{\nabla} u)_\theta r d\theta + (\vec{\nabla} u)_\phi r \sin \theta d\phi$$

and we demand that this hold for any choice of dr , $d\theta$, and $d\phi$. Thus,

$$(\vec{\nabla}u)_r = \frac{\partial u}{\partial r}, \quad (\vec{\nabla}u)_\theta = \frac{1}{r} \frac{\partial u}{\partial \theta}, \quad (\vec{\nabla}u)_\phi = \frac{1}{r \sin \theta} \frac{\partial u}{\partial \phi},$$

from which we find

$$\boxed{\vec{\nabla} = \hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi}}$$

Divergence

The divergence $\vec{\nabla} \cdot \vec{A}$ is carried out taking into account, once again, that the unit vectors themselves are functions of the coordinates. Thus, we have

$$\vec{\nabla} \cdot \vec{A} = \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \cdot (A_r \hat{r} + A_\theta \hat{\theta} + A_\phi \hat{\phi})$$

where the derivatives must be taken *before* the dot product so that

$$\begin{aligned} \vec{\nabla} \cdot \vec{A} &= \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \cdot \vec{A} \\ &= \hat{r} \cdot \frac{\partial \vec{A}}{\partial r} + \frac{\hat{\theta}}{r} \cdot \frac{\partial \vec{A}}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \cdot \frac{\partial \vec{A}}{\partial \phi} \\ &= \hat{r} \cdot \left(\frac{\partial A_r}{\partial r} \hat{r} + \frac{\partial A_\theta}{\partial r} \hat{\theta} + \frac{\partial A_\phi}{\partial r} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial r} + A_\theta \frac{\partial \hat{\theta}}{\partial r} + A_\phi \frac{\partial \hat{\phi}}{\partial r} \right) \\ &\quad + \frac{\hat{\theta}}{r} \cdot \left(\frac{\partial A_r}{\partial \theta} \hat{r} + \frac{\partial A_\theta}{\partial \theta} \hat{\theta} + \frac{\partial A_\phi}{\partial \theta} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial \theta} + A_\theta \frac{\partial \hat{\theta}}{\partial \theta} + A_\phi \frac{\partial \hat{\phi}}{\partial \theta} \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \cdot \left(\frac{\partial A_r}{\partial \phi} \hat{r} + \frac{\partial A_\theta}{\partial \phi} \hat{\theta} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial \phi} + A_\theta \frac{\partial \hat{\theta}}{\partial \phi} + A_\phi \frac{\partial \hat{\phi}}{\partial \phi} \right) \end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned} \vec{\nabla} \cdot \vec{A} &= \hat{r} \cdot \left(\frac{\partial A_r}{\partial r} \hat{r} + \frac{\partial A_\theta}{\partial r} \hat{\theta} + \frac{\partial A_\phi}{\partial r} \hat{\phi} + 0 + 0 + 0 \right) \\ &\quad + \frac{\hat{\theta}}{r} \cdot \left(\frac{\partial A_r}{\partial \theta} \hat{r} + \frac{\partial A_\theta}{\partial \theta} \hat{\theta} + \frac{\partial A_\phi}{\partial \theta} \hat{\phi} + A_r \hat{\theta} + A_\theta (-\hat{r}) + 0 \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \cdot \left(\frac{\partial A_r}{\partial \phi} \hat{r} + \frac{\partial A_\theta}{\partial \phi} \hat{\theta} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + A_r \sin \theta \hat{\phi} + A_\theta \cos \theta \hat{\phi} + A_\phi \left[-(\hat{r} \sin \theta + \hat{\theta} \cos \theta) \right] \right) \\ &= \left(\frac{\partial A_r}{\partial r} \right) + \left(\frac{1}{r} \frac{\partial A_\theta}{\partial \theta} + \frac{A_r}{r} \right) + \left(\frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi} + \frac{A_r}{r} + \frac{A_\theta \cos \theta}{r \sin \theta} \right) \\ &= \left(\frac{\partial A_r}{\partial r} + \frac{2A_r}{r} \right) + \left(\frac{1}{r} \frac{\partial A_\theta}{\partial \theta} + \frac{A_\theta \cos \theta}{r \sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi} \\ &\boxed{\vec{\nabla} \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}} \end{aligned}$$

Curl

The curl $\vec{\nabla} \times \vec{A}$ is also carried out taking into account that the unit vectors themselves are functions of the coordinates. Thus, we have

$$\vec{\nabla} \times \vec{A} = \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \times (A_r \hat{r} + A_\theta \hat{\theta} + A_\phi \hat{\phi})$$

where the derivatives must be taken *before* the cross product so that

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \times \vec{A} \\ &= \hat{r} \times \frac{\partial \vec{A}}{\partial r} + \frac{\hat{\theta}}{r} \times \frac{\partial \vec{A}}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \times \frac{\partial \vec{A}}{\partial \phi} \\ &= \hat{r} \times \left(\frac{\partial A_r}{\partial r} \hat{r} + \frac{\partial A_\theta}{\partial r} \hat{\theta} + \frac{\partial A_\phi}{\partial r} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial r} + A_\theta \frac{\partial \hat{\theta}}{\partial r} + A_\phi \frac{\partial \hat{\phi}}{\partial r} \right) \\ &\quad + \frac{\hat{\theta}}{r} \times \left(\frac{\partial A_r}{\partial \theta} \hat{r} + \frac{\partial A_\theta}{\partial \theta} \hat{\theta} + \frac{\partial A_\phi}{\partial \theta} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial \theta} + A_\theta \frac{\partial \hat{\theta}}{\partial \theta} + A_\phi \frac{\partial \hat{\phi}}{\partial \theta} \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \times \left(\frac{\partial A_r}{\partial \phi} \hat{r} + \frac{\partial A_\theta}{\partial \phi} \hat{\theta} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial \phi} + A_\theta \frac{\partial \hat{\theta}}{\partial \phi} + A_\phi \frac{\partial \hat{\phi}}{\partial \phi} \right) \end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \hat{r} \times \left(\frac{\partial A_r}{\partial r} \hat{r} + \frac{\partial A_\theta}{\partial r} \hat{\theta} + \frac{\partial A_\phi}{\partial r} \hat{\phi} + 0 + 0 + 0 \right) \\ &\quad + \frac{\hat{\theta}}{r} \times \left(\frac{\partial A_r}{\partial \theta} \hat{r} + \frac{\partial A_\theta}{\partial \theta} \hat{\theta} + \frac{\partial A_\phi}{\partial \theta} \hat{\phi} + A_r \hat{\theta} + A_\theta (-\hat{r}) + 0 \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \times \left(\frac{\partial A_r}{\partial \phi} \hat{r} + \frac{\partial A_\theta}{\partial \phi} \hat{\theta} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + A_r \sin \theta \hat{\phi} + A_\theta \cos \theta \hat{\phi} + A_\phi \left[-(\hat{r} \sin \theta + \hat{\theta} \cos \theta) \right] \right) \\ &= \left(\frac{\partial A_\theta}{\partial r} \hat{\phi} - \frac{\partial A_\phi}{\partial r} \hat{\theta} \right) + \left(-\frac{1}{r} \frac{\partial A_r}{\partial \theta} \hat{\phi} + \frac{1}{r} \frac{\partial A_\phi}{\partial \theta} \hat{r} + \frac{A_\theta}{r} \hat{\phi} \right) \\ &\quad + \left(\frac{1}{r \sin \theta} \frac{\partial A_r}{\partial \phi} \hat{\theta} - \frac{1}{r \sin \theta} \frac{\partial A_\theta}{\partial \phi} \hat{r} - \frac{A_\phi}{r} \hat{\theta} + \frac{A_\phi \cos \theta}{r \sin \theta} \hat{r} \right) \\ &= \hat{r} \left(\frac{1}{r} \frac{\partial A_\phi}{\partial \theta} - \frac{1}{r \sin \theta} \frac{\partial A_\theta}{\partial \phi} + \frac{A_\phi \cos \theta}{r \sin \theta} \right) \\ &\quad + \hat{\theta} \left(-\frac{\partial A_\phi}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{A_\phi}{r} \right) \\ &\quad + \hat{\phi} \left(\frac{\partial A_\theta}{\partial r} - \frac{1}{r} \frac{\partial A_r}{\partial \theta} + \frac{A_\theta}{r} \right) \end{aligned}$$

$$\boxed{\vec{\nabla} \times \vec{A} = \frac{\hat{r}}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] + \frac{\hat{\theta}}{r \sin \theta} \left[\frac{\partial A_r}{\partial \phi} - \sin \theta \frac{\partial}{\partial r} (r A_\phi) \right] + \frac{\hat{\phi}}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right]}$$

Laplacian

The Laplacian is a scalar operator that can be determined from its definition as

$$\begin{aligned}\nabla^2 u &= \vec{\nabla} \cdot (\vec{\nabla} u) = \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \cdot \left(\hat{r} \frac{\partial u}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial u}{\partial \phi} \right) \\ &= \hat{r} \cdot \frac{\partial}{\partial r} \left(\hat{r} \frac{\partial u}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial u}{\partial \phi} \right) \\ &\quad + \frac{\hat{\theta}}{r} \cdot \frac{\partial}{\partial \theta} \left(\hat{r} \frac{\partial u}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial u}{\partial \phi} \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \cdot \frac{\partial}{\partial \phi} \left(\hat{r} \frac{\partial u}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial u}{\partial \phi} \right)\end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned}\nabla^2 u &= \hat{r} \cdot \left(\hat{r} \frac{\partial^2 u}{\partial r^2} - \frac{\hat{\theta}}{r^2} \frac{\partial u}{\partial \theta} + \frac{\hat{\theta}}{r} \frac{\partial^2 u}{\partial \theta \partial r} - \frac{\hat{\phi}}{r^2 \sin \theta} \frac{\partial u}{\partial \phi} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial^2 u}{\partial \phi \partial r} \right) \\ &\quad + \frac{\hat{\theta}}{r} \cdot \left(\hat{\theta} \frac{\partial u}{\partial r} + \hat{r} \frac{\partial^2 u}{\partial r \partial \theta} - \frac{\hat{r}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\theta}}{r} \frac{\partial^2 u}{\partial \theta^2} - \frac{\hat{\phi} \cos \theta}{r \sin^2 \theta} \frac{\partial u}{\partial \phi} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial^2 u}{\partial \phi \partial \theta} \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \cdot \left(\hat{\phi} \sin \theta \frac{\partial u}{\partial r} + \hat{r} \frac{\partial^2 u}{\partial r \partial \phi} + \frac{\hat{\phi} \cos \theta}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\theta}}{r} \frac{\partial^2 u}{\partial \theta \partial \phi} - \frac{\hat{r} \sin \theta + \hat{\theta} \cos \theta}{r \sin \theta} \frac{\partial u}{\partial \phi} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial^2 u}{\partial \phi^2} \right) \\ &= \left(\frac{\partial^2 u}{\partial r^2} \right) + \left(\frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \right) + \left(\frac{1}{r} \frac{\partial u}{\partial r} + \frac{\cos \theta}{r^2 \sin \theta} \frac{\partial u}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} \right) \\ &= \left(\frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r} \right) + \left(\frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\cos \theta}{r^2 \sin \theta} \frac{\partial u}{\partial \theta} \right) + \left(\frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} \right) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}\end{aligned}$$

Thus, the Laplacian operator can be written as

$$\boxed{\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}}$$

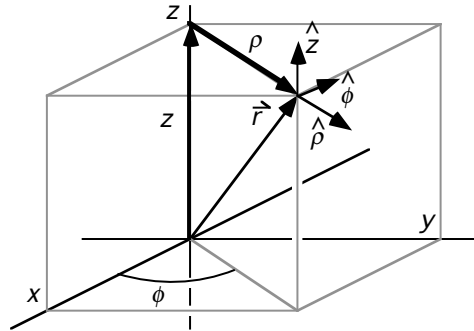
Cylindrical Coordinates

Transforms

The forward and reverse coordinate transformations are

$$\begin{aligned}\rho &= \sqrt{x^2 + y^2} & x &= \rho \cos \phi \\ \phi &= \arctan(y, x) & y &= \rho \sin \phi \\ z &= z & z &= z\end{aligned}$$

where we *formally* take advantage of the *two argument* arctan function to eliminate quadrant confusion.



Unit Vectors

The unit vectors in the cylindrical coordinate system are functions of position. It is convenient to express them in terms of the *cylindrical* coordinates and the unit vectors of the *rectangular* coordinate system which are *not* themselves functions of position.

$$\begin{aligned}\hat{\rho} &= \frac{\vec{\rho}}{\rho} = \frac{x\hat{x} + y\hat{y}}{\rho} = \hat{x} \cos \phi + \hat{y} \sin \phi \\ \hat{\phi} &= \hat{z} \times \hat{\rho} = -\hat{x} \sin \phi + \hat{y} \cos \phi \\ \hat{z} &= \hat{z}\end{aligned}$$

Variations of unit vectors with the coordinates

Using the expressions obtained above it is easy to derive the following handy relationships:

$$\begin{aligned}\frac{\partial \hat{\rho}}{\partial \rho} &= 0 & \frac{\partial \hat{\phi}}{\partial \rho} &= 0 & \frac{\partial \hat{z}}{\partial \rho} &= 0 \\ \frac{\partial \hat{\rho}}{\partial \phi} &= -\hat{x} \sin \phi + \hat{y} \cos \phi = \hat{\phi} & \frac{\partial \hat{\phi}}{\partial \phi} &= -\hat{x} \cos \phi - \hat{y} \sin \phi = -\hat{\rho} & \frac{\partial \hat{z}}{\partial \phi} &= 0 \\ \frac{\partial \hat{\rho}}{\partial z} &= 0 & \frac{\partial \hat{\phi}}{\partial z} &= 0 & \frac{\partial \hat{z}}{\partial z} &= 0\end{aligned}$$

Path increment

We will have many uses for the path increment $d\vec{r}$ expressed in cylindrical coordinates:

$$\begin{aligned}d\vec{r} &= d(\rho\hat{\rho} + z\hat{z}) = \hat{\rho}d\rho + \rho d\hat{\rho} + \hat{z}dz + z d\hat{z} \\ &= \hat{\rho}d\rho + \rho \left(\frac{\partial \hat{\rho}}{\partial \rho} d\rho + \frac{\partial \hat{\rho}}{\partial \phi} d\phi + \frac{\partial \hat{\rho}}{\partial z} dz \right) + \hat{z}dz + z \left(\frac{\partial \hat{z}}{\partial \rho} d\rho + \frac{\partial \hat{z}}{\partial \phi} d\phi + \frac{\partial \hat{z}}{\partial z} dz \right) \\ &= \hat{\rho}d\rho + \hat{\phi}\rho d\phi + \hat{z}dz\end{aligned}$$

Time derivatives of the unit vectors

We will also have many uses for the time derivatives of the unit vectors expressed in cylindrical coordinates:

$$\begin{aligned}\dot{\hat{\rho}} &= \frac{\partial \hat{\rho}}{\partial \rho} \dot{\rho} + \frac{\partial \hat{\rho}}{\partial \phi} \dot{\phi} + \frac{\partial \hat{\rho}}{\partial z} \dot{z} = \hat{\phi} \dot{\phi} \\ \dot{\hat{\phi}} &= \frac{\partial \hat{\phi}}{\partial \rho} \dot{\rho} + \frac{\partial \hat{\phi}}{\partial \phi} \dot{\phi} + \frac{\partial \hat{\phi}}{\partial z} \dot{z} = -\hat{\rho} \dot{\phi} \\ \dot{\hat{z}} &= \frac{\partial \hat{z}}{\partial \rho} \dot{\rho} + \frac{\partial \hat{z}}{\partial \phi} \dot{\phi} + \frac{\partial \hat{z}}{\partial z} \dot{z} = 0\end{aligned}$$

Velocity and Acceleration

The velocity and acceleration of a particle may be expressed in cylindrical coordinates by taking into account the associated rates of change in the unit vectors:

$$\begin{aligned}\vec{v} = \dot{\vec{r}} &= \dot{\hat{\rho}}\rho + \hat{\rho}\dot{\rho} + \dot{\hat{z}}z + \hat{z}\dot{z} = \hat{\phi}\dot{\phi}\rho + \hat{z}\dot{z} \\ \vec{v} &= \hat{\rho}\dot{\rho} + \hat{\phi}\rho\dot{\phi} + \hat{z}\dot{z} \\ \vec{a} = \dot{\vec{v}} &= \dot{\hat{\rho}}\dot{\rho} + \hat{\rho}\ddot{\rho} + \dot{\hat{\phi}}\rho\dot{\phi} + \hat{\phi}\dot{\rho}\dot{\phi} + \hat{\phi}\rho\ddot{\phi} + \dot{\hat{z}}\dot{z} + \hat{z}\ddot{z} \\ &= \hat{\phi}\dot{\phi}\dot{\rho} + \hat{\rho}\ddot{\rho} - \hat{\rho}\rho\dot{\phi}^2 + \hat{\phi}\dot{\rho}\dot{\phi} + \hat{\phi}\rho\ddot{\phi} + \hat{z}\ddot{z} \\ \vec{a} &= \hat{\rho}(\ddot{\rho} - \rho\dot{\phi}^2) + \hat{\phi}(\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}) + \hat{z}\ddot{z}\end{aligned}$$

The del operator from the definition of the gradient

Any (static) scalar field u may be considered to be a function of the cylindrical coordinates ρ , ϕ , and z . The value of u changes by an infinitesimal amount du when the point of observation is changed by $d\vec{r}$. That change may be determined from the partial derivatives as

$$du = \frac{\partial u}{\partial \rho} d\rho + \frac{\partial u}{\partial \phi} d\phi + \frac{\partial u}{\partial z} dz.$$

But we also define the gradient in such a way as to obtain the result

$$du = \vec{\nabla} u \cdot d\vec{r}$$

Therefore,

$$\frac{\partial u}{\partial \rho} d\rho + \frac{\partial u}{\partial \phi} d\phi + \frac{\partial u}{\partial z} dz = \vec{\nabla} u \cdot d\vec{r}$$

or, in cylindrical coordinates,

$$\frac{\partial u}{\partial \rho} d\rho + \frac{\partial u}{\partial \phi} d\phi + \frac{\partial u}{\partial z} dz = (\vec{\nabla} u)_\rho d\rho + (\vec{\nabla} u)_\phi \rho d\phi + (\vec{\nabla} u)_z dz$$

and we demand that this hold for any choice of $d\rho$, $d\phi$ and dz . Thus,

$$(\vec{\nabla} u)_\rho = \frac{\partial u}{\partial \rho}, \quad (\vec{\nabla} u)_\phi = \frac{1}{\rho} \frac{\partial u}{\partial \phi}, \quad (\vec{\nabla} u)_z = \frac{\partial u}{\partial z},$$

from which we find

$$\vec{\nabla} = \hat{\rho} \frac{\partial}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial}{\partial \phi} + \hat{z} \frac{\partial}{\partial z}$$

Divergence

The divergence $\vec{\nabla} \cdot \vec{A}$ is carried out taking into account, once again, that the unit vectors themselves are functions of the coordinates. Thus, we have

$$\vec{\nabla} \cdot \vec{A} = \left(\hat{\rho} \frac{\partial}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial}{\partial \phi} + \hat{z} \frac{\partial}{\partial z} \right) \cdot (A_\rho \hat{\rho} + A_\phi \hat{\phi} + A_z \hat{z})$$

where the derivatives must be taken *before* the dot product so that

$$\begin{aligned} \vec{\nabla} \cdot \vec{A} &= \left(\hat{\rho} \frac{\partial}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial}{\partial \phi} + \hat{z} \frac{\partial}{\partial z} \right) \cdot \vec{A} \\ &= \hat{\rho} \cdot \frac{\partial \vec{A}}{\partial \rho} + \frac{\hat{\phi}}{\rho} \cdot \frac{\partial \vec{A}}{\partial \phi} + \hat{z} \cdot \frac{\partial \vec{A}}{\partial z} \\ &= \hat{\rho} \cdot \left(\frac{\partial A_\rho}{\partial \rho} \hat{\rho} + \frac{\partial A_\phi}{\partial \rho} \hat{\phi} + \frac{\partial A_z}{\partial \rho} \hat{z} + A_\rho \frac{\partial \hat{\rho}}{\partial \rho} + A_\phi \frac{\partial \hat{\phi}}{\partial \rho} + A_z \frac{\partial \hat{z}}{\partial \rho} \right) \\ &\quad + \frac{\hat{\phi}}{\rho} \cdot \left(\frac{\partial A_\rho}{\partial \phi} \hat{\rho} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + \frac{\partial A_z}{\partial \phi} \hat{z} + A_\rho \frac{\partial \hat{\rho}}{\partial \phi} + A_\phi \frac{\partial \hat{\phi}}{\partial \phi} + A_z \frac{\partial \hat{z}}{\partial \phi} \right) \\ &\quad + \hat{z} \cdot \left(\frac{\partial A_\rho}{\partial z} \hat{\rho} + \frac{\partial A_\phi}{\partial z} \hat{\phi} + \frac{\partial A_z}{\partial z} \hat{z} + A_\rho \frac{\partial \hat{\rho}}{\partial z} + A_\phi \frac{\partial \hat{\phi}}{\partial z} + A_z \frac{\partial \hat{z}}{\partial z} \right) \end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned} \vec{\nabla} \cdot \vec{A} &= \hat{\rho} \cdot \left(\frac{\partial A_\rho}{\partial \rho} \hat{\rho} + \frac{\partial A_\phi}{\partial \rho} \hat{\phi} + \frac{\partial A_z}{\partial \rho} \hat{z} + 0 + 0 + 0 \right) \\ &\quad + \frac{\hat{\phi}}{\rho} \cdot \left(\frac{\partial A_\rho}{\partial \phi} \hat{\rho} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + \frac{\partial A_z}{\partial \phi} \hat{z} + A_\rho \hat{\phi} - A_\phi \hat{\rho} + 0 \right) \\ &\quad + \hat{z} \cdot \left(\frac{\partial A_\rho}{\partial z} \hat{\rho} + \frac{\partial A_\phi}{\partial z} \hat{\phi} + \frac{\partial A_z}{\partial z} \hat{z} + 0 + 0 + 0 \right) \\ &= \left(\frac{\partial A_\rho}{\partial \rho} \right) + \left(\frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{A_\rho}{\rho} \right) + \left(\frac{\partial A_z}{\partial z} \right) \\ &= \left(\frac{\partial A_\rho}{\partial \rho} + \frac{A_\rho}{\rho} \right) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z} \end{aligned}$$

$$\boxed{\vec{\nabla} \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (A_\rho \rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}}$$

Curl

The curl $\vec{\nabla} \times \vec{A}$ is also carried out taking into account that the unit vectors themselves are functions of the coordinates. Thus, we have

$$\vec{\nabla} \times \vec{A} = \left(\hat{\rho} \frac{\partial}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial}{\partial \phi} + \hat{z} \frac{\partial}{\partial z} \right) \times (A_\rho \hat{\rho} + A_\phi \hat{\phi} + A_z \hat{z})$$

where the derivatives must be taken *before* the cross product so that

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \left(\hat{\rho} \frac{\partial}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial}{\partial \phi} + \hat{z} \frac{\partial}{\partial z} \right) \times \vec{A} \\ &= \hat{\rho} \times \frac{\partial \vec{A}}{\partial \rho} + \frac{\hat{\phi}}{\rho} \times \frac{\partial \vec{A}}{\partial \phi} + \hat{z} \times \frac{\partial \vec{A}}{\partial z} \\ &= \hat{\rho} \times \left(\frac{\partial A_\rho}{\partial \rho} \hat{\rho} + \frac{\partial A_\phi}{\partial \rho} \hat{\phi} + \frac{\partial A_z}{\partial \rho} \hat{z} + A_\rho \frac{\partial \hat{\rho}}{\partial \rho} + A_\phi \frac{\partial \hat{\phi}}{\partial \rho} + A_z \frac{\partial \hat{z}}{\partial \rho} \right) \\ &\quad + \frac{\hat{\phi}}{\rho} \times \left(\frac{\partial A_\rho}{\partial \phi} \hat{\rho} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + \frac{\partial A_z}{\partial \phi} \hat{z} + A_\rho \frac{\partial \hat{\rho}}{\partial \phi} + A_\phi \frac{\partial \hat{\phi}}{\partial \phi} + A_z \frac{\partial \hat{z}}{\partial \phi} \right) \\ &\quad + \hat{z} \times \left(\frac{\partial A_\rho}{\partial z} \hat{\rho} + \frac{\partial A_\phi}{\partial z} \hat{\phi} + \frac{\partial A_z}{\partial z} \hat{z} + A_\rho \frac{\partial \hat{\rho}}{\partial z} + A_\phi \frac{\partial \hat{\phi}}{\partial z} + A_z \frac{\partial \hat{z}}{\partial z} \right) \end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \hat{\rho} \times \left(\frac{\partial A_\rho}{\partial \rho} \hat{\rho} + \frac{\partial A_\phi}{\partial \rho} \hat{\phi} + \frac{\partial A_z}{\partial \rho} \hat{z} + 0 + 0 + 0 \right) \\ &\quad + \frac{\hat{\phi}}{\rho} \times \left(\frac{\partial A_\rho}{\partial \phi} \hat{\rho} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + \frac{\partial A_z}{\partial \phi} \hat{z} + A_\rho \hat{\phi} - A_\phi \hat{\rho} + 0 \right) \\ &\quad + \hat{z} \times \left(\frac{\partial A_\rho}{\partial z} \hat{\rho} + \frac{\partial A_\phi}{\partial z} \hat{\phi} + \frac{\partial A_z}{\partial z} \hat{z} + 0 + 0 + 0 \right) \\ &= \left(\frac{\partial A_\phi}{\partial \rho} \hat{z} - \frac{\partial A_z}{\partial \rho} \hat{\phi} \right) + \left(-\frac{1}{\rho} \frac{\partial A_\rho}{\partial \phi} \hat{z} + \frac{1}{\rho} \frac{\partial A_z}{\partial \phi} \hat{\rho} + \frac{A_\phi}{\rho} \hat{z} \right) \\ &\quad + \left(\frac{\partial A_\rho}{\partial z} \hat{\phi} - \frac{\partial A_\phi}{\partial z} \hat{\rho} \right) \\ &= \hat{\rho} \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \hat{\phi} \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) + \hat{z} \left(\frac{\partial A_\phi}{\partial \rho} + \frac{A_\phi}{\rho} - \frac{1}{\rho} \frac{\partial A_\rho}{\partial \phi} \right) \end{aligned}$$

$$\vec{\nabla} \times \vec{A} = \hat{\rho} \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \hat{\phi} \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) + \hat{z} \left(\frac{1}{\rho} \frac{\partial}{\partial \rho} (A_\phi \rho) - \frac{1}{\rho} \frac{\partial A_\rho}{\partial \phi} \right)$$

Laplacian

The Laplacian is a scalar operator that can be determined from its definition as

$$\begin{aligned}
 \nabla^2 u &= \vec{\nabla} \cdot (\vec{\nabla} u) = \left(\hat{\rho} \frac{\partial}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial}{\partial \phi} + \hat{z} \frac{\partial}{\partial z} \right) \cdot \left(\hat{\rho} \frac{\partial u}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial u}{\partial \phi} + \hat{z} \frac{\partial u}{\partial z} \right) \\
 &= \hat{\rho} \cdot \frac{\partial}{\partial \rho} \left(\hat{\rho} \frac{\partial u}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial u}{\partial \phi} + \hat{z} \frac{\partial u}{\partial z} \right) \\
 &\quad + \frac{\hat{\phi}}{\rho} \cdot \frac{\partial}{\partial \phi} \left(\hat{\rho} \frac{\partial u}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial u}{\partial \phi} + \hat{z} \frac{\partial u}{\partial z} \right) \\
 &\quad + \hat{z} \cdot \frac{\partial}{\partial z} \left(\hat{\rho} \frac{\partial u}{\partial \rho} + \frac{\hat{\phi}}{\rho} \frac{\partial u}{\partial \phi} + \hat{z} \frac{\partial u}{\partial z} \right)
 \end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned}
 \nabla^2 u &= \hat{\rho} \cdot \left(\hat{\rho} \frac{\partial^2 u}{\partial \rho^2} - \frac{\hat{\phi}}{\rho^2} \frac{\partial u}{\partial \phi} + \frac{\hat{\phi}}{\rho} \frac{\partial^2 u}{\partial \phi \partial \rho} + \hat{z} \frac{\partial^2 u}{\partial z \partial \rho} \right) \\
 &\quad + \frac{\hat{\phi}}{\rho} \cdot \left(\hat{\phi} \frac{\partial u}{\partial \rho} + \hat{\rho} \frac{\partial^2 u}{\partial \rho \partial \phi} - \frac{\hat{\rho}}{\rho} \frac{\partial u}{\partial \phi} + \frac{\hat{\phi}}{\rho} \frac{\partial^2 u}{\partial \phi^2} + \hat{z} \frac{\partial^2 u}{\partial z \partial \phi} \right) \\
 &\quad + \hat{z} \cdot \left(\hat{\rho} \frac{\partial^2 u}{\partial \rho \partial z} + \frac{\hat{\phi}}{\rho} \frac{\partial^2 u}{\partial \phi \partial z} + \hat{z} \frac{\partial^2 u}{\partial z^2} \right) \\
 &= \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} \\
 &= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}
 \end{aligned}$$

Thus, the Laplacian operator can be written as

$$\boxed{\nabla^2 = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2}}$$

Chapter 2 特征值问题

2.1 1

Chapter 3 傅里叶变换

3.1 傅里叶级数

魏尔斯特拉斯定理告诉我们任意关于 x 的连续函数都可以展开成 x 的代数多项式。这个结论可以推广到 n 个自变量的情形：

Theorem 3.1.1 (Generalized Stone-Weierstrass Theorem): 如果函数 $f(x_1, x_2, \dots, x_n)$ 是区间 $[a_i, b_i]^n$ 上的连续函数，那么 $f(x_1, x_2, \dots, x_n)$ 可以展开成多项式 $x_1^{k_1} x_2^{k_2} \dots x_n^{k_n}$, k_i 是非负整数。

由 $x_1 = x = r \cos \theta, x_2 = y = r \sin \theta$ ，我们得到了极坐标下展开一个二元函数的方法：

$$f(r, \theta) = \sum_{m,k=0}^{\infty} a_{mk} x^k y^m = \sum_{m,k=0}^{\infty} a_{m,k} r^{m+k} \cos^k \theta \sin^m \theta \quad (3.1)$$

如果我们令 $r = 1$ （或者其他常数）就得到了一个只关于 θ 的函数：

$$f(\theta) = \sum_{n=-\infty}^{\infty} F_n e^{in\theta} = F_0 + \sum_{n=1}^{\infty} (A_n \cos \theta + B_n \sin \theta) \quad (3.2)$$

这是一个周期为 2π 的周期性函数，它可以表示 $[-\pi, \pi]$ 上的连续函数（或者分段连续函数，准确地讲是满足狄利克雷条件的函数）。利用正交性和归一性，不难求出 [1]：

$$F_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\theta} f(\theta) d\theta \quad (3.3)$$

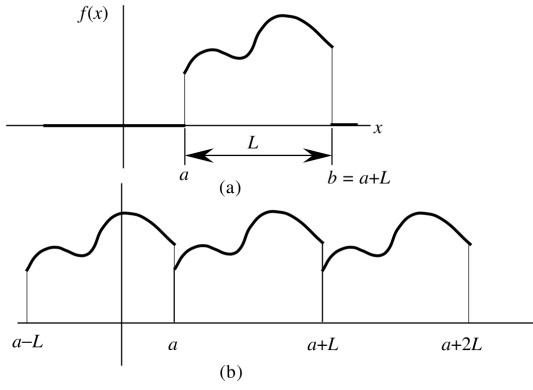
但是，我们感兴趣的函数并不总是定义在 $[-\pi, \pi]$ 上。更一般的，对于定义在 (a, b) 上，周期为 $L=(b-a)$ 的函数 $f(x)$ ，通过换元

$$\theta = \frac{2\pi}{L}(x - a - \frac{L}{2})$$

可以得到

$$f(x) = \sum_{n=-\infty}^{\infty} F_n e^{2\pi i n x / L} \quad (3.4)$$

$$F_n = \frac{1}{L} \int_a^b e^{-2\pi i n x / L} f(x) dx \quad (3.5)$$



(a) The function we want to represent. (b) The Fourier series representation of the function

3.2 傅里叶变换

傅里叶级数可以展开任意周期性函数 $F(x)$, 但是物理学中遇到的大多数函数并不是周期性的。一种简单的想法是使周期 $L \rightarrow \infty$, 此时 $\Delta\omega = \frac{2\pi}{L} \rightarrow 0$ 。[2] 我们直接给出傅里叶变换的定义

$$F[f(x)] = G(\omega) = \int_{-\infty}^{\infty} f(x)e^{-i\omega x} dx \quad (3.6)$$

$$F^{-1}[G(\omega)] = f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega)e^{i\omega x} d\omega \quad (3.7)$$

我们称公式3.6为函数 $f(x)$ 的傅里叶变换, $F[f(x)] = G(\omega)$ 称为 $f(x)$ 的像函数. 公式3.7称为 $G(\omega)$ 的傅里叶逆变换, $f(x)$ 称为 $G(\omega)$ 的像原函数.

上面的定义可以推广到三维:

$$G(\omega) = \int_{-\infty}^{\infty} f(r)e^{-i\omega \cdot r} dr \quad (3.8)$$

$$f(r) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega)e^{i\omega \cdot r} d\omega \quad (3.9)$$

其中:

$$\begin{cases} \omega = \omega_1 \mathbf{e}_1 + \omega_2 \mathbf{e}_2 + \omega_3 \mathbf{e}_3 \\ \mathbf{r} = x \mathbf{e}_1 + y \mathbf{e}_2 + z \mathbf{e}_3 \\ d\mathbf{r} = dx dy dz \\ f(\mathbf{r}) = f(x, y, z) \end{cases}$$

3.2.1 傅里叶变换的性质

- 线性性质: $F[\alpha f_1 + \beta f_2] = \alpha F[f_1] + \beta F[f_2]$.
- 延迟性质: $F[e^{i\omega_0 x} f(x)] = G(\omega - \omega_0)$.
- 位移性质: $F[f(x - x_0)] = e^{-i\omega x_0} F[f(x)]$.
- 相似性质: $F[f(ax)] = \frac{1}{|a|} G(\frac{\omega}{a})$.
- 微分性质: $F[f^{(n)}(x)] = (i\omega)^n F[f(x)]$, 正是由于傅里叶变换的微分性质, 引出了求解微分方程的积分变换法.

- 积分性质: $F[\int_{x_0}^x f(\xi)d\xi] = \frac{1}{i\omega}F[f(x)]$.
- 卷积性质: $f_1(x) * f_2(x) = \int_{-\infty}^{\infty} f_1(x)f_2(x-\xi)d\xi$ 称为卷积, 卷积满足交换律、结合律、分配律。对函数和像函数:

$$F[f_1(x) * f_2(x)] = F[f_1(x)] \cdot F[f_2(x)] \quad (\text{卷积定理}) \quad (3.10)$$

$$F[f_1(x) \cdot f_2(x)] = \frac{1}{2\pi} F[f_1(x)] * F[f_2(x)] \quad (3.11)$$

Chapter 4 数学物理方程

4.1 达朗贝尔法

4.2 分离变量法

4.3 积分变换法

Chapter 5 格林函数

Chapter 6 一些物理学中常用的函数

Chapter 7 张量

Chapter 8 物理学中的群论

Chapter 9 经典力学

Chapter 10 量子力学基础

本章主要介绍量子力学中的基本概念和数学工具，包括狄拉克符号、表象理论、角动量、全同粒子等，关于量子力学的诠释和历史，将会在自然哲学部分中介绍。

参考书目：

- 《量子力学导论》，格里菲斯，很经典的入门书籍，有中文版；
- 《量子力学》，科恩-塔诺季，值得收藏的一本书，大师之作，详略得当，附录非常全面；
- 《量子力学原理》，王正行，这本书花费了很多篇幅在狄拉克符号上，适合对此感到困惑的初学者；
- 《高等量子力学》，喀兴林，表象理论和全同粒子介绍的比较详细。

10.1 量子力学的基本假设

1. 描写微观系统状态的数学量是希尔伯特空间 (态空间) 中的矢量 (态矢量 $|\psi\rangle$)，相差一个复数因子的两个矢量，描写同一状态。
2. 每一个可测量的物理量 $\mathcal{A}$ 都可以用希尔伯特空间中的一个厄米算符 A 描述，因此之后我们统一使用大写字母标记，这个算符是一个观察算符；
3. 物理量所能取的值 a_i ，只能是相应算符 A 的本征值之一，因为 A 是厄米的，因此测量结果总是实数；

$$A|a_i\rangle = a_i|a_i\rangle \quad \langle a_i|a_j\rangle = \delta_{ij} \quad (10.1)$$

4. 物理量 A 在状态 $|\psi\rangle$ 中取各值 a_i 的概率，与态矢量 $|\psi\rangle$ 按 A 的归一化本征矢量 $|a_i\rangle$ 的展开式中 $|a_i\rangle$ 的系数 c_i 的复平方成正比。若 $|\psi\rangle$ 是归一化的，那么测量物理量 A 得到 a_i 的概率是 $|c_i|^2$ 。因此可以写出 A 在状态 $|\psi\rangle$ 的平均值 $\langle A \rangle$ ：

$$|\psi\rangle = \sum_i |a_i\rangle c_i = \sum_i |a_i\rangle \langle a_i|\psi\rangle \quad (10.2)$$

$$\langle A \rangle = \frac{\sum_i c_i^* c_i a_i}{\sum_i c_i^* c_i} = \frac{\sum_i \langle \psi|a_i\rangle \langle a_i|\psi\rangle a_i}{1} = \langle \psi|A|\psi\rangle \quad (10.3)$$

5. 微观系统中每个粒子的直角坐标下的位置算符 X_i ($i=1, 2, 3$)，与相应的正则动量算符 P ，有下列对易关系：

$$[X_i, X_j] = 0 \quad [P_i, P_j] = 0 \quad [X_i, P_j] = i\hbar\delta_{ij} \quad \text{其中 } \hbar = \frac{h}{2\pi} \quad (10.4)$$

而不同粒子间的所有算符相互对易 $[A_i, B_j] = 0$ 。

6. 微观系统的状态随时间演化的方式是 (含时) 薛定谔方程:

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H|\psi\rangle \quad (10.5)$$

只要给出了初始状态 $|\psi(0)\rangle$ 就足以知道此后任意时刻的状态 $|\psi(t)\rangle$ 。在演化过程中没有任何不确定性, 只在测量时发生不可预测的跃变。

7. 描写全同粒子系统的态矢量, 对于任意一对粒子的对调, 是对称的 (对调前后完全相同) 或反对称的 (对调前后差一个负号), 服从前者的粒子称为玻色子, 服从后者的粒子称为费米子。

10.2 谐振子和矩阵方法

10.3 角动量

10.4 微扰论

10.5 变分法

10.6 散射

Chapter 11 量子力学 II

11.1 全同粒子和二次量子化

11.2 路径积分

11.3 狄拉克方程

Chapter 12 狭义相对论

12.1 相对性原理

参考系 为了描述自然界中发生的过程, 我们必须要有个参考系, 它包含了一个坐标系和一个固定在参考系中的钟.

惯性系 如果一个不受外界作用的物体, 在参考系中静止或者做匀速直线运动, 这样的参考系称为惯性系. 狭义相对论中涉及的参考系全部都是惯性系. 如果存在一个基本惯性系, 相对于此惯性系做匀速直线运动的另一坐标系也是一个惯性系.

这不是显然的, 因为它涉及到了速度的参考系变换: 如果在 K 系中存在速度为常数 u 的物体和速度为常数 v 的参考系 K' , 那么在 K' 系中物体的速度 u' 也是一个常数 (所以 K' 是惯性系). 然而在确定 K' 是惯性系之前, 我们不能应用任何关于惯性系的原理和物理定律. 因此最后一句话不是一个推论, 而是一个同样重要的假设或者定义: 牛顿第一定律成立的参考系是一个基本惯性系, 相对于惯性系静止或者匀速运动的参考系称为惯性系. 事实上后半句才是惯性系的定义, 牛顿第一定律 (关于惯性系性质的公设) 的引入帮助我们确定了最初的基本惯性系. "不受外界作用" 可以理解为孤立的、远离其它粒子的, 从而避免了依赖牛顿第一定律进行判断引发 "先有鸡还是先有蛋" 的问题.

狭义相对论基本假设

- **相对性原理**: 所有的自然定律在所有惯性系中是相同的.
- **光速不变原理**: 物体之间的相互作用不是瞬时的, 从一个物体向另一个物体发出的相互作用称做 "信号", 信号传播速度 (光速) 在不同惯性系中是相同的 (相对性原理).

$$c = 2.998 \times 10^8 m/s$$

12.1.1 经典力学的时空观

经典力学中的空间观念是相对的, 当我们说不同时刻的两个事件发生在同一地点时, 必须指明参考系才有意义. 比如火车上一只苍蝇飞起, 盘旋一会儿后落回了原来的位置, 就是相对火车上的乘客而言的, 但是对大地参考系却不是这样. 经典力学的时间观念是绝对的, 纯粹均匀地流逝而不因参考系的选择而变化.

12.1.2 相对论的时空观

同时的相对性 考虑一个匀速行驶的长度为 $2L$ 的车厢, 车厢顶部挂有一个灯泡. 现在打开灯泡, 在车厢参考系中光线以相同的速度 c 向车厢两边传播, 同时到达车厢的 (a)(b) 两端. 但是在地面参考系中, 光线从灯泡初始时刻的位置发出后, 以相同的速度 c (而不是 $c-v$ 和 $c+v$) 向车厢两端传播, 与此同时车厢在向右运动, 光线相对于车厢左端 (b) 的速度是 $c+v$, 而相对于 (a) 的速度是 $c-v$, 因此光线将先到达 (b) 端. 注意, 光速不变是相对于参考系的, 而不是参考系中被描述的其它物体的.



图 12.1: 同时的相对性

钟慢效应

尺缩效应

12.2 间隔

事件 指的是时间和空间中的一个点, 在四维坐标系下用世界点 (t, x, y, z) 来刻画。所有事件的集合构成了我们的时空。然而, 需要强调的是事件本身与坐标系选择无关, 在不同坐标系下事件的坐标是不同的。而世界线 (worldline) 指的是一个点粒子在时空中的轨迹。 s_{12} 称为间隔, 不同参考系中的间隔是相等的 (这样的量称为洛伦兹不变量, LI), 利用间隔的这种性质可以方便地求出不同惯性系中坐标和时间的变换关系。我们采用高能物理常用的西海岸约定 $(+ - - -)$, 而在天体物理中常用东海岸约定 $(- + + +)$ 。

$$s_{12}^2 = (c(t_2 - t_1))^2 - (x_2 - x_1)^2 - (y_2 - y_1)^2 - (z_2 - z_1)^2 = c^2 t_{12}^2 - l_{12}^2 \quad (12.1)$$

s_{12} 是虚数, 即 $s_{12}^2 > 0$ 的间隔称为**类时间隔**, 类时间隔两端的事件可以在某个参考系中同一点发生:

$$s_{12}^2 = c^2 t_{12}^2 - l_{12}^2 = c^2 t_{12}'^2 \quad l_{12}'^2 = 0 \quad (12.2)$$

类似的还有**类空间隔**, $s_{12}^2 < 0$, 如果两个事件的间隔是类空间隔, 那么他们可能在某个参考系中同时发生:

$$s_{12}^2 = c^2 t_{12}^2 - l_{12}^2 = -c^2 t_{12}'^2 \quad t_{12}'^2 = 0$$

还有**类光间隔**, 如果 $s_{12}^2 = 0$, 这与光的传播类似。

12.3 洛伦兹变换

洛伦兹变换 从一个惯性系中坐标 (x, y, z, t) 到另一个惯性系中 (x', y', z', t') 的变换公式称为洛伦兹变换。如果我们在有三个空间实轴 x, y, z 和一个虚轴 ict 四维闵可夫斯基空间考虑参考系变换的问题, 不难发现, 间隔对应两个世界点在四维坐标系中的距离。参考系变换就是将世界点的在一套坐标系中的坐标变换到另外一套坐标轴中对应的坐标, 保持"距离"不变的坐标的变换只有两种: 旋转和平移。而平移只涉及空间坐标原点和零时刻的选择, 所以我们所要求的变换对应坐标轴的转动。

假设 K' 系相对 K 系沿 x 方向以速度 v 运动:

$$\begin{pmatrix} \cosh \psi & \sinh \psi \\ \sinh \psi & \cosh \psi \end{pmatrix} \begin{pmatrix} x' \\ t' \end{pmatrix} = \begin{pmatrix} x \\ t \end{pmatrix} \quad (12.3)$$

在伪欧几里得几何的转动矩阵中三角函数被双曲函数取代。考虑 K' 系的原点 $x'=0$ 在 K 系中的坐标 x :

$$x = ct' \sinh \psi \quad ct = ct' \cosh \psi \quad (12.4)$$

于是我们得到

$$\tanh \psi = \frac{x}{ct} = \frac{v}{c} \quad (12.5)$$

由此得到了从运动系 K' 到静止系 K 的变换公式, 亦即著名的**洛伦兹变换**:

$$x = \gamma(x' + vt') \quad t = \gamma(t' + \frac{v}{c^2}x') \quad y = y' \quad z = z' \quad (12.6)$$

其中 $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$, $\beta = \frac{v}{c}$. 反之:

$$x' = \gamma(x - vt) \quad t' = \gamma(t - \frac{v}{c^2}x) \quad y' = y \quad z' = z \quad (12.7)$$

我们还可以得到速度 $\mathbf{u} = \frac{d\mathbf{r}}{dt}$ 的变换公式:

$$\begin{aligned} dx &= \gamma(dx' + v dt') \quad dt = \gamma(dt' + \frac{v}{c^2}dx') \quad dy = dy' \quad dz = dz' \\ u_x &= \frac{u'_x}{\sqrt{1 + u'_x \frac{v}{c^2}}} \quad u_y = \frac{1/\gamma u'_y}{\sqrt{1 + u'_x \frac{v}{c^2}}} \quad u_z = \frac{1/\gamma u'_z}{\sqrt{1 + u'_x \frac{v}{c^2}}} \end{aligned} \quad (12.8)$$

12.4 四维矢量

用以下方法表示洛伦兹变换时, 形式会简化许多:

$$x^\mu = \{ct, x, y, z\} \quad x_\mu = \{ct, -x, -y, -z\} \quad (12.9)$$

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} \quad (12.10)$$

或者用张量的形式写为:

$$x'^\mu = \sum_{\nu=0}^3 \Lambda_\nu^\mu x^\nu = \Lambda_\nu^\mu x^\nu \quad (12.11)$$

我们引入**爱因斯坦求和规则**: 当出现重复指标时就意味着求和, 而把求和号省略掉. 每对指标中必须一个上标, 另一个为下标. 这种" 傀" 指标用起来非常方便.

四维矢量 在洛伦兹变换下与 x^μ 有相同变换规则的一组四分量矢量称为 4-矢量:

$$a'^\mu = \Lambda_\nu^\mu a^\nu \quad (12.12)$$

a^μ 称为逆变矢量, $a_{\mu\nu}$ 称为协变矢量.

$$a^\mu = (a^0, a^1, a^2, a^3) = g^{\mu\nu} a_\nu \quad (12.13)$$

$$a_\mu = (a^0, -a^1, -a^2, -a^3) = g_{\mu\nu} a^\nu \quad (12.14)$$

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (12.15)$$

在引入了四维矢量之后, 间隔也可以改写为:

$$s = x^\mu x_\mu \quad (12.16)$$

它是一个**四维标量积**, 正如普通的点积在三维坐标转动下是不变的, 四维标量积在洛伦兹变换下也是不变的, 称为**洛伦兹不变量 (Lorentz Invariant, LI)**

Chapter 13 固体物理基础

13.1 从波函数到宏观物理量

凝聚态物理的起点是多粒子的薛定谔方程, $\Psi(\{\mathbf{r}_i\}, t)$ 是反对称的多粒子波函数, 简单起见我们暂时不考虑自旋。

$$i\hbar \frac{d\Psi(\{\mathbf{r}_i\}, t)}{dt} = \hat{H}\Psi(\{\mathbf{r}_i\}, t) \quad (13.1)$$

哈密顿算符是位置算符、动量算符（以及时间和系统所处外部环境）的函数，其函数形式与经典力学的哈密顿量相同，包含了系统与相互作用相关的全部信息。

$$\hat{H} = -\frac{\hbar^2}{2m_e} \sum_i \nabla_i^2 - \frac{\hbar^2}{2M_I} \sum_I \nabla_I^2 - \sum_{i,I} \frac{Z_I e^2}{|\mathbf{r}_i - \mathbf{R}_I|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{2} \sum_{I \neq J} \frac{Z_I Z_J e^2}{|\mathbf{R}_I - \mathbf{R}_J|} \quad (13.2)$$

其中 Z_I 和 M_I 是原子核的电荷数和质量。

13.1.1 电子密度

电子密度算符:

$$\hat{n}(\mathbf{r}) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \quad (13.3)$$

期望值:

$$n(\mathbf{r}) = \frac{\langle \Psi | \hat{n}(\mathbf{r}) | \Psi \rangle}{\langle \Psi | \Psi \rangle} = N \frac{\int d^3r_2 d^3r_3 \cdots d^3r_N |\Psi(\mathbf{r}, r_2, r_3 \cdots r_N)|^2}{\int d^3r_1 d^3r_2 d^3r_3 \cdots d^3r_N |\Psi(r_1, r_2, r_3 \cdots r_N)|^2} \quad (13.4)$$

13.1.2 能量

忽略掉原子核的动能，系统的哈密顿量可以分为以下几部分：

$$\hat{H} = \hat{T} + \hat{V}_{\text{ext}} + \hat{V}_{\text{int}} + E_{II} \quad (13.5)$$

它们分别对应于公式电子动能、电子与核的相互作用、电子之间的相互作用、原子核之间的相互作用。 $\hat{H}$ 的期望值也可以拆成这几项：

$$E \equiv \langle \hat{H} \rangle = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \langle \hat{T} \rangle + \langle \hat{V}_{\text{int}} \rangle + \int d^3r V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) + E_{II} \quad (13.6)$$

如果将能量 E 看做是波函数 Ψ 的泛函，那么哈密顿量的本征态就是使能量取得极值的鞍点。我们用拉格朗日乘子法处理公式13.6，得到的结果是定态薛定谔方程：

$$\hat{H}|\Psi\rangle = E|\Psi\rangle \quad (13.7)$$

13.2 晶体结构

- 理想的晶体是由相同的**原子团**（注意是原子团而不是原子）在空间中**无限重复**排列构成的，这样的原子团称为**基元**（*basis*）。
- 在研究晶体的结构时，这些原子团可以抽象为几何点，所有点的集合称为**晶格**（*lattice*）。
- 由于晶体具有平移对称性，可以通过平移 $u_1\vec{a}_1 + u_2\vec{a}_2 + u_3\vec{a}_3$ 使每一个格点移动到完全等价的另一个格点。如果以上的操作可以遍历全部格点，那么这个晶格就称为初基晶格， $\vec{a}_i$ 称为**初基平移矢量**（*primitive translation vector*）。基元内原子 j 的中心位置也可以用 $\vec{a}$ 来表示：

$$\mathbf{r}_j = x_j\mathbf{a}_1 + y_j\mathbf{a}_2 + z_j\mathbf{a}_3$$

初基平移矢量的选取不是唯一的。我们往往用初基平移矢量来定义晶轴。

- 体积最小的晶胞称为原胞。初基平移矢量所确定的平行六面体 ($\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3$) 是原胞的一种选取方法，因为初基平移矢量的选取不唯一，原胞选取也不唯一。
- **维格纳-赛茨原胞**：原胞的另一种取法，把一个格点和相邻格点用直线连接起来，做连线的垂直平分线或者垂直平分面，围成的最小体积。

13.3 倒空间

周期性的函数如晶体中的势能、电子密度等可以展开成傅里叶级数，以电子密度为例：

$$n(\mathbf{r}) = \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}} \quad (13.8)$$

$$n_{\mathbf{G}} = \frac{1}{V_c} \int_{cell} n(\vec{r}) e^{-i\mathbf{G} \cdot \mathbf{r}} d\vec{r} \quad (13.9)$$

$\mathbf{G}$ 不是任意的，因为 $n(\mathbf{r})$ 具有平移不变性：

$$n(\mathbf{r} + \vec{T}) = \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}} e^{i\mathbf{G} \cdot \mathbf{T}} = n(\mathbf{r}) \quad (13.10)$$

我们定义倒格子的轴矢量：

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3} \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_2 \cdot \mathbf{a}_3 \times \mathbf{a}_1} \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_3 \cdot \mathbf{a}_1 \times \mathbf{a}_3} \quad (13.11)$$

倒空间每个初基矢量和实空间的两个轴矢量正交：

$$\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi \delta_{ij} \quad (13.12)$$

当 $\mathbf{G} = v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + v_3\mathbf{b}_3$ (v_i 是整数) 时, $\exp(i\mathbf{G} \cdot \mathbf{T}) = 1$, 因此 $\mathbf{G} = v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + v_3\mathbf{b}_3$ 就是波矢被允许的取值范围。

13.3.1 布里渊区

布里渊区的定义是倒格子空间中的维格纳-赛茨原胞，它的边界是服从衍射条件的波矢 $\mathbf{k}$ ：

$$2\mathbf{k} \cdot \mathbf{G} = G^2 \quad (13.13)$$

倒格子的中央晶胞称为第一布里渊区。

13.4 自由电子气

13.5 固体能带理论

13.5.1 波恩-冯卡门边界条件

13.5.2 近自由电子近似

近自由电子近似将晶体中离子实对电子的作用, 看过是一个周期势场微扰. 一个具有平移操作不变性的周期性势场可以展开成倒格矢的傅里叶级数:

$$U(x) = \sum_G U_G e^{iGx} \quad (13.14)$$

13.5.3 布洛赫定理

平移算符和哈密顿算符对易, 二者具有相同的本征函数:

$$\hat{H}\psi(r) = E\psi(r) \quad \hat{T}(r)\psi(r) = C(r)\psi(r) \quad (13.15)$$

平移操作满足交换律 $\hat{T}(r_1)\hat{T}(r_2) = \hat{T}(r_2)\hat{T}(r_1) = \hat{T}(r_1 + r_2)$, 因此布拉维平移群是阿贝尔群, 根据舒尔引理, 平移算符的本征值 $C(r)$ 构成一维表示. 由于群乘法规则 $\hat{T}(r_1)\hat{T}(r_2) = \hat{T}(r_1 + r_2)$, 本征值需要满足: $C(r_1)C(r_2) = C(r_1 + r_2)$, 唯一连续解是指数函数:

$$C(r) = e^{ik \cdot r} \quad (13.16)$$

于是 $\hat{T}(a)\psi(r) = \psi(r + a) = e^{ik \cdot a}\psi(r)$ (注意不是级数). 为了满足这一关系, 波函数必须包含一个随位置变化的相位因子 $e^{ik \cdot r}$, 以抵消平移带来的相位变化. 将这部分显式地提取出来:

$$\psi(r) = e^{ik \cdot r}u(r) \rightarrow \psi(r + a) = e^{ik \cdot (r+a)}u(r + a) = e^{ik \cdot a}e^{ik \cdot r}u(r) \quad (13.17)$$

因此 $u(r)$ 是周期为 a 的函数:

$$u(r + a) = u(r) \quad (13.18)$$

13.5.4 电子在周期性势场的波动方程

考虑一个晶格常数为 a , 首尾相连 (波恩-冯卡门边界条件), 周长为 L 的一维晶格圆环. 单电子近似下, 晶体中电子的波动方程可以写为以下形式:

$$\left(\frac{1}{2m}p^2 + U(x)\right)\psi(x) = \epsilon\psi(x) \quad (13.19)$$

我们将单电子波函数 $\psi(x)$ 展开,

$$\psi(x) = \sum_k C_k e^{ikx} \quad (13.20)$$

波矢 k 需要满足 $k = \frac{2\pi n}{L}$ (n 是整数), 因为 $\psi(x)$ 是周期为 L 的周期性函数 (参阅傅里叶级数). 布洛赫定理对 k 提出了额外的要求:

$$\begin{cases} \psi(x) = e^{ikx} \sum_{k'} C_{k'} e^{i(k'-k)x} \\ \psi(x+a) = e^{ik(x+a)} \sum_{k'} C_{k'} e^{i(k'-k)x} e^{i(k'-k)a} \end{cases} \rightarrow e^{i(k'-k)a} = 1 \rightarrow k' = k + G$$

因此一旦波矢 k 出现在 $\psi(x)$ 的展开式中, 那么其它展开项只能是符合 $k+G$ 的波矢. 将势能和波函数的级数代入, 得到:

$$\sum_k \frac{\hbar^2 k^2}{2m} C_k e^{ikx} + \sum_G \sum_k U_G C_k e^{i(k+G)x} = \epsilon \sum_k C_k e^{ikx} \quad (13.21)$$

方程两端各个 k 的傅里叶分量系数相等:

$$(\lambda_k - \epsilon) C_k + \sum_G U_G C_{k-G} = 0 \quad \text{其中} \quad \lambda_k = \frac{\hbar^2 k^2}{2m} \quad (13.22)$$

上式称为**中心方程**.

13.6 附录: 三维空间的十四种晶格

13.7 附录: 晶体衍射条件

Chapter 14 粒子物理基础

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